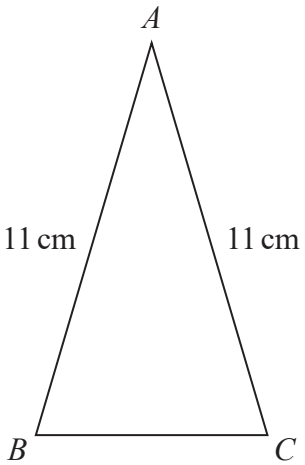
20 ABC is an isosceles triangle.



The area of the triangle is 42.5 cm^2

Work out the size of angle ABC.
Give your answer correct to 1 decimal place.

Area = ½ × 11 × 11 × sin A = 42.5

1 mark
Use the two equal sides and their included apex angle.

sin A = 85/121

1 mark
Rearrange, then use inverse sine.

For the acute apex shown, A = $\sin^{-1}(85/121)$

A = 44.626261...°

Equal base angles: B = C = (180° − A) ÷ 2

Angle ABC = 67.686869...° ≈ 67.7°

1 mark
Official intended branch: the acute apex illustrated.

67.7 °

(Total for Question 20 is 3 marks)

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